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Method for manufacturing silicon photonic device and silicon photonic device thereof

Abstract

A method for manufacturing an integrated circuit device is provided. The method includes: providing a photonic structure including an insulating structure and an optical coupler embedded in the insulating structure; and removing a portion of the insulating structure to expose a coupling surface of the optical coupler and form a light reflective structure corresponding to the coupling surface.

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Background/Summary

BACKGROUND
(1) As for silicon photonic devices such as optical transceivers, vertical coupling can accomplish wafer-level testing. A grating coupler has been applied to perform the vertical coupling. However, the wavelength selectivity of the grating coupler significantly limits its bandwidth, causing it difficult to achieve broadband testing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(3) FIG. **2** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(4) FIG. **3** illustrates a cross-sectional view of one or more stages of an example of a method for

manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(5) FIG. **4** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(6) FIG. **5** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(7) FIG. **6** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(8) FIG. **7** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(9) FIG. **8** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(10) FIG. **9** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(11) FIG. **10** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(12) FIG. **11** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(13) FIG. **12** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(14) FIG. **13** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(15) FIG. **14** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(16) FIG. **15** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(17) FIG. **16** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(18) FIG. **17** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(19) FIG. **18** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(20) FIG. **19** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.

(21) FIG. **20** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(22) FIG. **21** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(23) FIG. **22** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(24) FIG. **23** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(25) FIG. **24** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(26) FIG. **25** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.
(27) FIG. **26** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(28) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(29) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “upper,” “on” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(30) As used herein, although the terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.
(31) Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements.

Also, as used herein, the terms “substantially,” “approximately” and “about” generally mean within a value or range that can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately” and “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

(32) FIG. **1** through FIG. **10** illustrate a method for manufacturing an integrated circuit device **1** according to some embodiments of the present disclosure. In some embodiments, the integrated circuit device **1** can be a high-speed transceiver.

(33) Referring to FIG. **1**, a photonic structure **10** is provided. The photonic structure **10** can be formed on a substrate **30**. The substrate **30** has a first surface **31** (e.g., a top surface) and a second surface **32** (e.g., a bottom surface) opposite to the first surface **31**. The photonic structure **10** can be formed on the second surface **32** of the substrate **30**. In some embodiments, the substrate **30** can include an oxide structure **35** covering the first surface **31**.

(34) The photonic structure **10** can also be referred to as “photonic die” or “P-die.” As shown in FIG. **1**, the photonic structure **10** includes an insulating structure **14**, an optical coupler **15**, a photodetector **16**, at least one circuit layer **17** in contact with the insulating structure **14**, a plurality of bonding pads **18** and a plurality of inner conductive vias **19**.

(35) In some embodiments, as shown in FIG. **1**, the insulating structure **14** can have a top surface **140** and include a plurality of insulating layers (including, for example, a first insulating layer **141**, a second insulating layer **142**, a third insulating layer **143**, a fourth insulating layer **144** and a fifth insulating layer **145**). The top surface **140** is substantially coplanar with the second surface **32** of the substrate **30**. The insulating layers (e.g., the first insulating layer **141**, the second insulating layer **142**, the third insulating layer **143**, the fourth insulating layer **144** and the fifth insulating layer **145**) are stacked on one another. For example, the first insulating layer **141** can be the topmost insulating layer and formed on the second surface **32** of the substrate **30**. A material of the insulating layers (e.g., the first insulating layer **141**, the second insulating layer **142**, the third insulating layer **143**, the fourth insulating layer **144** and the fifth insulating layer **145**) can be, for example, buried oxide.

(36) The optical coupler **15** can be, for example, a waveguide coupler. The optical coupler **15** is embedded in the insulating structure **14**. That is, the optical coupler **15** is in contact with the insulating structure **14**. As shown in FIG. **1**, the optical coupler **15** is embedded in the second insulating layer **142** and on a bottom surface of the first insulating layer **141**. The optical coupler **15** is configured to receive a light signal. In some embodiments, the optical coupler **15** can have a top surface **151**, a bottom surface **152** opposite to the top surface **151** and a coupling surface **153** (or an edge) extending between the top surface **151** and the bottom surface **152**.

(37) The photodetector **16** is embedded in the insulating structure **14** and spaced apart from the optical coupler **15**. As shown in FIG. **1**, the photodetector **16** is embedded in the second insulating layer **142** and on the bottom surface of the first insulating layer **141**. The photodetector **16** is configured to convert the light signal into an electric signal.

(38) The at least one circuit layer **17** can include a first circuit layer **171** and a second circuit layer

172. The first circuit layer **171** is formed on a bottom surface of the second insulating layer **142** and covered by the third insulating layer **143**. The second circuit layer **172** is formed on a bottom surface of the third insulating layer **143** and covered by the fourth insulating layer **144**. A material of the circuit layer **17** (including, for example, the first circuit layer **171** and the second circuit layer **172**) can include, for example, copper, another conductive metal, or an alloy thereof.

(39) The bonding pads **18** are formed on a bottom surface of the fourth insulating layer **144** and exposed from a bottom surface of the fifth insulating layer **145**. The inner conductive vias **19** can include at least one first inner conductive via **191**, at least one second inner conductive via **192** and at least one third inner conductive via **193**. The first inner conductive via **191** extends through the second insulating layer **142** and is disposed between the photodetector **16** and the first circuit layer **171** for electrically connecting the photodetector **16** and the first circuit layer **171**. In some embodiments, the first inner conductive via **191** and the first circuit layer **171** can be formed concurrently and integrally. The second inner conductive via **192** extends through the third insulating layer **143** and is disposed between the first circuit layer **171** and the second circuit layer **172** for electrically connecting the first circuit layer **171** and the second circuit layer **172**. In some embodiments, the second inner conductive via **192** and the second circuit layer **172** can be formed concurrently and integrally. The third inner conductive via **193** extends through the fourth insulating layer **144** and is disposed between the second circuit layer **172** and the bonding pads **18** for electrically connecting the second circuit layer **172** and the bonding pads **18**. In some embodiments, the third inner conductive via **193** and the bonding pads **18** can be formed concurrently and integrally.

(40) Referring to FIG. **2** and FIG. **3**, an electronic structure **20** is provided, and the photonic structure **10** and the electronic structure **20** are bonded together. The electronic structure **20** can also be referred to as “electronic die” or “E-die.” As shown in FIG. **2**, the electronic structure **20** includes a substrate **23**, an insulating structure **24**, at least one circuit layer **27** in contact with the insulating structure **24**, a plurality of bonding pads **28** and a plurality of inner conductive vias **29**. The substrate **23** can be, for example, a logic substrate.

(41) In some embodiments, as shown in FIG. **2**, the insulating structure **24** can include a plurality of insulating layers (including, for example, a first insulating layer **241**, a second insulating layer **242**, a third insulating layer **243** and a fourth insulating layer **244**). The insulating layers (e.g., the first insulating layer **241**, the second insulating layer **242**, the third insulating layer **243** and the fourth insulating layer **244**) are stacked on one another. For example, the first insulating layer **241** can be the bottommost insulating layer and formed on a top surface of the substrate **23**. A material of the insulating layers (e.g., the first insulating layer **241**, the second insulating layer **242**, the third insulating layer **243** and the fourth insulating layer **244**) can be, for example, buried oxide.

(42) The at least one circuit layer **27** can include a first circuit layer **271** and a second circuit layer **272**. The first circuit layer **271** is formed on a top surface of the first insulating layer **241** and covered by the second insulating layer **242**. The second circuit layer **272** is formed on a top surface of the second insulating layer **242** and covered by the third insulating layer **243**. A material of the circuit layer **27** (including, for example, the first circuit layer **271** and the second circuit layer **272**) can include, for example, copper, another conductive metal, or an alloy thereof.

(43) The bonding pads **28** are formed on a top surface of the third insulating layer **243** and exposed from a top surface of the fourth insulating layer **244**. The inner conductive vias **29** can include at least one first inner conductive via **291** and at least one second inner conductive via **292**. The first inner conductive via **291** extends through the second insulating layer **242** and is disposed between the first circuit layer **271** and the second circuit layer **272** for electrically connecting the first circuit layer **271** and the second circuit layer **272**. In some embodiments, the first inner conductive via **291** and the second circuit layer **272** can be formed concurrently and integrally. The second inner conductive via **292** extends through the third insulating layer **243** and is disposed between the second circuit layer **272** and the bonding pads **28** for electrically connecting the second circuit layer

272 and the bonding pads **28**. In some embodiments, the second inner conductive via **292** and the bonding pads **28** can be formed concurrently and integrally. As shown in FIG. **3**, the bonding pads **18** of the photonic structure **10** and the bonding pads **28** of the electronic structure **20** are bonded together.

(44) Referring to FIG. **4**, the substrate **30** is removed to expose the insulating structure **14** (including, for example, the first insulating layer **141**, the second insulating layer **142**, the third insulating layer **143**, the fourth insulating layer **144** and the fifth insulating layer **145**).

(45) Referring to FIG. **5**, a first photoresist layer **51** is formed on the top surface **140** (e.g., a top surface of the first insulating layer **141**) of the insulating structure **14** to cover a portion of the top surface **140** by, for example, coating. In some embodiments, as shown in FIG. **5**, the insulating structure **14** can define a first portion **146** and a second portion **147** below the first portion **146**. The first photoresist layer **51** can expose the first portion **146** of the insulating structure **14**.

(46) Referring to FIG. **6**, the first portion **146** of the insulating structure **14** is removed to expose the coupling surface **153** (or the edge) of the optical coupler **15** and the second portion **147** of the insulating structure **14** by, for example, etching. In some embodiments, the optical coupler **15** can also be referred to as “edge coupler.” As shown in FIG. **6**, the coupling surface **153** (or the edge) is exposed from and substantially coplanar with a lateral surface **14**S of the insulating structure **14**.

(47) Referring to FIG. **7**, a second photoresist layer **52** is formed on a top surface of the second portion **147** of the insulating structure **14** to cover a portion of the top surface of the second portion **147** by, for example, coating. In some embodiments, as shown in FIG. **7**, the second portion **147** can define an inner portion **1471** and an outer portion **1472** opposite to the inner portion **1471**. The second photoresist layer **52** can cover the outer portion **1472** and expose the inner portion **1471**.

(48) Referring to FIG. **8**, the inner portion **1471** of the second portion **147** is removed to form a rib portion **148** (i.e., the outer portion **1472**) on the insulating structure **14** by, for example, etching. Then, the first photoresist layer **51** and the second photoresist layer **52** are removed. As shown in FIG. **8**, the rib portion **148** has a corner portion **149**.

(49) Referring to FIG. **9**, the rib portion **148** is shaped to form a light reflective structure **60** by, for example, plasma hitting. The light reflective structure **60** corresponds to the coupling surface **153** of the optical coupler **15**. In some embodiments, the light reflective structure **60** can include an inclined reflective surface **62**. The plasma can hit the corner portion **149** of the rib portion **148** to form the inclined reflective surface **62** on the rib portion **148**. In some embodiments, an angle θ between the inclined reflective surface **62** and a lateral surface **1483** of the rib portion **148** can be greater than or equal to 90 degrees. The inclined reflective surface **62** can be a mirror-like surface. In some embodiments, as shown in FIG. **9**, the inclined reflective surface **62** can be lower than the optical coupler **15**. In some embodiments, the inclined reflective surface **62** can be lower than the top surface **151** of the optical coupler **15**. In some embodiments, the inclined reflective surface **62** can be lower than the bottom surface **152** of the optical coupler **15**.

(50) As shown in FIG. **9**, the photonic structure **10**, the electronic structure **20** and the light reflective structure **60** (including, for example, the inclined reflective surface **62**) can constitute an integrated circuit device **1**.

(51) Referring to FIG. **10**, a light testing signal **70** from an optical fiber **80** is coupled into the coupling surface **153** of the optical coupler **15** through the light reflective structure **60** (including, for example, the inclined reflective surface **62**). Then, the light testing signal **70** is coupled to the photodetector **16** from one end of the optical coupler **15**. The photodetector **16** can convert the light testing signal **70** into an electric testing signal, and output the electric testing signal to the electronic structure **20** to complete the test.

(52) The method of the present disclosure can be applied in an integrated circuit device including the photonic structure **10** as illustrated above; however, the disclosure is not limited thereto. As shown in the embodiments illustrated in FIG. **1** through FIG. **10**, the light reflective structure **60** (including, for example, the inclined reflective surface **62**) is directly formed on the photonic

structure **10**. An optical path of the light testing signal **70** can be changed from vertical coupling to edge coupling through the light reflective structure **60** (including, for example, the inclined reflective surface **62**). Additionally, the optical coupler **15** can be used as the edge coupler, and the light reflective structure **60** (including, for example, the inclined reflective surface **62**) can increase the bandwidth of the edge coupler (i.e., the optical coupler **15**) to achieve broadband testing.

(53) FIG. **11** illustrates a method for manufacturing an integrated circuit device **1***a* according to some embodiments of the present disclosure. The initial stages of the illustrated process are the same as, or similar to, the stages illustrated in FIG. **1** through FIG. **9**. FIG. **11** depicts a stage subsequent to that depicted in FIG. **9**.

(54) Referring to FIG. **11**, a light reflective film **90** is formed or disposed on the inclined reflective surface **62**. A material of the light reflective film **90** can include metal, metal oxide, or metal nitride. The metal can be, for example, aluminum (Al), gold (Au), copper (Cu), or tantalum (Ta). The metal oxide can be, for example, titanium dioxide (TiO_2). The metal nitride can be, for example, Tantalum nitride (TaN). In some embodiments, the light reflective film **90** can be a distributed Bragg reflector (DBR) film.

(55) FIG. **12** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device according to some embodiments of the present disclosure. The stage illustrated in FIG. **12** is the same as, or similar to, the stage illustrated in FIG. **1**, except for a structure of the photonic structure **10***a*. In some embodiments, as shown in FIG. **12**, an upper light reflector **81** can be formed in the insulating structure **14** (e.g., the first insulating layer **141**) and above the optical coupler **15**, and a lower light reflector **82** can be formed in the insulating structure **14** (e.g., the second insulating layer **142**) and below the optical coupler **15**. The upper light reflector **81** is configured to guide the light signal leaking from the top surface **151** of the optical coupler **15** back into the optical coupler **15**, and the lower light reflector **82** is configured to guide the light signal leaking from the bottom surface **152** of the optical coupler **15** back into the optical coupler **15**, lead to increased light coupling efficiency.

(56) FIG. **13** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device **1***b* according to some embodiments of the present disclosure. The stage illustrated in FIG. **13** is the same as, or similar to, the stage illustrated in FIG. **9**, except for a configuration of the inclined reflective surface **62***a*. In some embodiments, as shown in FIG. **13**, an angle θ_a between the inclined reflective surface **62***a* and the lateral surface **1483** of the rib portion **148** can be different from the angle θ of FIG. **9**. The inclined reflective surface **62***a* can have a highest point P_1 and a lowest point P_2. As shown in FIG. **13**, an elevation of the highest point P_1 can be lower than an elevation of the top surface **151** of the optical coupler **15**. An elevation of the lowest point P_2 can be higher than an elevation of the bottom surface **152** of the optical coupler **15**. In addition, a vertical distance Da between the elevation of the top surface **151** of the optical coupler **15** and the elevation of the lowest point P_2 can be less than a thickness t of the optical coupler **15**. In some embodiments, an incident angle of the light testing signal **70** (FIG. **10**) can be adjusted based on the angle θ_a.

(57) FIG. **14** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device **1***c* according to some embodiments of the present disclosure. The stage illustrated in FIG. **14** is the same as, or similar to, the stage illustrated in FIG. **13**, except for a configuration of the inclined reflective surface **62***b*. In some embodiments, as shown in FIG. **14**, an angle θ_b between the inclined reflective surface **62***b* and the lateral surface **1483** of the rib portion **148** can be different from the angle θ_a of FIG. **13**. The elevation of the highest point P_1 can be higher than the elevation of the top surface **151** of the optical coupler **15**. In addition, a vertical distance Db between the elevation of the top surface **151** of the optical coupler **15** and the elevation of the lowest point P_2 can be less than the thickness t of the optical coupler **15**. In some embodiments, the incident angle of the light testing signal **70** (FIG. **10**) can be adjusted based on the angle θ_b.

(58) FIG. **15** illustrates a cross-sectional view of one or more stages of an example of a method for manufacturing an integrated circuit device **1***d* according to some embodiments of the present disclosure. The stage illustrated in FIG. **15** is the same as, or similar to, the stage illustrated in FIG. **13**, except for a configuration of the inclined reflective surface **62***c*. In some embodiments, as shown in FIG. **15**, an angle θ.sub.c between the inclined reflective surface **62***c* and the lateral surface **1483** of the rib portion **148** can be different from the angle θ.sub.a of FIG. **13**. The elevation of the lowest point P.sub.2 can be lower than the elevation of the bottom surface **152** of the optical coupler **15**. In addition, a vertical distance De between the elevation of the top surface **151** of the optical coupler **15** and the elevation of the lowest point P.sub.2 can be greater than the thickness t of the optical coupler **15**. In some embodiments, the incident angle of the light testing signal **70** (FIG. **10**) can be adjusted based on the angle θ.sub.c.
(59) FIG. **16** through FIG. **25** illustrates a method for manufacturing an integrated circuit device **1***e* according to some embodiments of the present disclosure. The stage illustrated in FIG. **16** is the same as, or similar to, the stage illustrated in FIG. **1**, except for an amount of the optical coupler **15** and an amount of the photodetector **16**. In some embodiments, as shown in FIG. **16**, the photonic structure **10***e* can include at least two optical couplers (including, for example, a first optical coupler **156** and a second optical coupler **157**) and at least two photodetectors (including, for example, a first photodetector **161** and a second photodetector **162**). The at least two optical couplers (e.g., the first optical coupler **156** and the second optical coupler **157**) are disposed side by side and spaced apart from each other. The first optical coupler **156** and the second optical coupler **157** of FIG. **16** can be the same as the optical coupler **15** of FIG. **1**. That is, the first optical coupler **156** can have a top surface **1561**, a bottom surface **1562** opposite to the top surface **1561** and a coupling surface **1563** (or an edge) extending between the top surface **1561** and the bottom surface **1562**. The second optical coupler **157** can have a top surface **1571**, a bottom surface **1572** opposite to the top surface **1571** and a coupling surface **1573** (or an edge) extending between the top surface **1571** and the bottom surface **1572**. The first photodetector **161** and the second photodetector **162** of FIG. **16** can be the same as the photodetector **16** of FIG. **1**. In addition, a configuration between the first optical coupler **156** and the first photodetector **161** and a configuration between the second optical coupler **157** and the second photodetector **162** can be the same as the configuration between the optical coupler **15** and the photodetector **16** of FIG. **1**.
(60) The stages illustrated in FIG. **17** through FIG. **19** are the same as, or similar to, the stages illustrated in FIG. **2** through FIG. **4**. In addition, the electronic structure **20***e* of FIG. **17** can be the same as the electronic structure **20** of FIG. **2**.
(61) Referring to FIG. **20**, a first photoresist layer **53** is formed on a top surface **140** (e.g., a top surface of the first insulating layer **141**) of the insulating structure **14** to cover a portion of the top surface **140** by, for example, coating. In some embodiments, as shown in FIG. **20**, the insulating structure **14** can define a first portion **146′** and a second portion **147′** below the first portion **146′**. The first photoresist layer **53** can define an opening **532** extending through the first photoresist layer **53**. The opening **532** can expose the first portion **146′** of the insulating structure **14**.
(62) Referring to FIG. **21**, the first portion **146′** of the insulating structure **14** is removed to expose the coupling surface **1563** (or the edge) of the first optical coupler **156**, the coupling surface **1573** (or the edge) of the second optical coupler **157** and the second portion **147′** of the insulating structure **14** by, for example, etching. In some embodiments, the first optical coupler **156** and the second optical coupler **157** can also be referred to as “edge coupler.”
(63) Referring to FIG. **22**, a second photoresist layer **54** is formed on a top surface of the second portion **147′** of the insulating structure **14** to cover a portion of the top surface of the second portion **147′** by, for example, coating. In some embodiments, as shown in FIG. **22**, the second portion **147′** can define an inner portion **1471′**, an outer portion **1472′** opposite to the inner portion **1471′** and an intermediate portion **1473′** between the inner portion **1471′** and the outer portion **1472′**. The second photoresist layer **54** can cover the intermediate portion **1473′** and expose the inner portion **1471′**

and the outer portion **1472**′.

(64) Referring to FIG. **23**, the inner portion **1471**′ and the outer portion **1472**′ of the second portion **147**′ are removed to form a rib portion **148**′ (i.e., the intermediate portion **1473**′) on the insulating structure **14** and between the at least two optical couplers (including, for example, the first optical coupler **156** and the second optical coupler **157**) by, for example, etching. Then, the first photoresist layer **53** and the second photoresist layer **54** are removed. As shown in FIG. **23**, the rib portion **148**′ has a first corner portion **1491** and a second corner portion **1492** opposite to the first corner portion **1491**.

(65) Referring to FIG. **24**, the rib portion **148**′ is shaped to form a light reflective structure **60***e* between the at least two optical couplers (e.g., between the first optical coupler **156** and the second optical coupler **157**) by, for example, plasma hitting. The light reflective structure **60***e* corresponds to the coupling surface **1563** of the first optical coupler **156** and the coupling surface **1573** of the second optical coupler **157**. In some embodiments, the light reflective structure **60***e* can include a first inclined reflective surface **64** corresponding to the coupling surface **1563** of the first optical coupler **156** and a second inclined reflective surface **65** corresponding to the coupling surface **1573** of the second optical coupler **157**. That is, an inclined direction of the first inclined reflective surface **64** is opposite to an inclined direction of the second inclined reflective surface **65**. The plasma can hit the first corner portion **1491** of the rib portion **148**′ to form the first inclined reflective surface **64** and hit the second corner portion **1492** of the rib portion **148**′ to form the second inclined reflective surface **65**. In some embodiments, an angle θ.sub.1 between the first inclined reflective surface **64** and a lateral surface **1483**′ of the rib portion **148**′ can be greater than 90 degrees. An angle θ.sub.2 between the second inclined reflective surface **65** and the lateral surface **1483**′ of the rib portion **148**′ can be greater than 90 degrees. In some embodiments, the angle θ.sub.1 can be different from the angle θ.sub.2. The first inclined reflective surface **64** and the second inclined reflective surface **65** can be a mirror-like surface. In some embodiments, as shown in FIG. **24**, the first inclined reflective surface **64** can be lower than the top surface **1561** of the first optical coupler **156**. The second inclined reflective surface **65** can be lower than the top surface **1571** of the second optical coupler **157**.

(66) As shown in FIG. **24**, the photonic structure **10***e*, the electronic structure **20***e* and the light reflective structure **60***e* (including, for example, the first inclined reflective surface **64** and the second inclined reflective surface **65**) can constitute an integrated circuit device **1***e*.

(67) Referring to FIG. **25**, a first light testing signal **71** from a first optical fiber **83** is coupled into the coupling surface **1563** of the first optical coupler **156** through the first inclined reflective surface **64** of the light reflective structure **60***e*. A second light testing signal **72** from a second optical fiber **84** is coupled into the coupling surface **1573** of the second optical coupler **157** through the second inclined reflective surface **65** of the light reflective structure **60***e*. Then, the first light testing signal **71** is coupled to the first photodetector **161** from one end of the first optical coupler **156**, and the second light testing signal **72** is coupled to the second photodetector **162** from one end of the second optical coupler **157**. The first photodetector **161** and the second photodetector **162** can respectively convert the first light testing signal **71** and the second light testing signal **72** into electric testing signals, and output the electric testing signals to the electronic structure **20***e* to complete the test.

(68) The method of the present disclosure can be applied in an integrated circuit device including the photonic structure **10***e* as illustrated above; however, the disclosure is not limited thereto. As shown in the embodiments illustrated in FIG. **16** through FIG. **25**, the light reflective structure **60***e* (including, for example, the first inclined reflective surface **64** and the second inclined reflective surface **65**) is formed between the at least two optical couplers (including, for example, the first optical coupler **156** and the second optical coupler **157**). Optical paths of the first light testing signal **71** and the second light testing signal **72** can be changed from vertical coupling to edge coupling through the light reflective structure **60***e* (including, for example, the first inclined

reflective surface **64** and the second inclined reflective surface **65**). Additionally, the first optical coupler **156** and the second optical coupler **157** can be used as edge couplers. The light reflective structure **60***e* (including, for example, the first inclined reflective surface **64** and the second inclined reflective surface **65**) can increase the bandwidth of the edge couplers (i.e., the first optical coupler **156** and the second optical coupler **157**) to achieve double-side broadband testing.
(69) FIG. **26** illustrates a method for manufacturing an integrated circuit device if according to some embodiments of the present disclosure. The initial stages of the illustrated process are the same as, or similar to, the stages illustrated in FIG. **16** through FIG. **24**. FIG. **26** depicts a stage subsequent to that depicted in FIG. **24**.
(70) Referring to FIG. **26**, a light reflective film **90***f* is formed or disposed on the first inclined reflective surface **64** and the second inclined reflective surface **65**. A material of the light reflective film **90***f* can include metal, metal oxide, or metal nitride. The metal can be, for example, aluminum (Al), gold (Au), copper (Cu), or tantalum (Ta). The metal oxide can be, for example, titanium dioxide (TiO.sub.2). The metal nitride can be, for example, Tantalum nitride (TaN). In some embodiments, the light reflective film **90***b* can be a distributed Bragg reflector (DBR) film.
(71) In accordance with some embodiments of the present disclosure, a method for manufacturing an integrated circuit device includes: providing a photonic structure including an insulating structure and an optical coupler embedded in the insulating structure; and removing a portion of the insulating structure to expose a coupling surface of the optical coupler and form a light reflective structure corresponding to the coupling surface.
(72) In accordance with some embodiments of the present disclosure, a method for manufacturing an integrated circuit device includes: providing a photonic structure including an insulating structure and at least two optical couplers embedded in the insulating structure and disposed side by side; and removing a portion of the insulating structure to form a light reflective structure between the at least two optical couplers.
(73) In accordance with some embodiments of the present disclosure, an integrated circuit device includes a photonic structure and a light reflective structure. The photonic structure includes an insulating structure and an optical coupler in contact with the insulating structure. The optical coupler has a coupling surface exposed from a lateral surface of the insulating structure. The light reflective structure is disposed on the photonic structure and corresponds to the coupling surface of the optical coupler.
(74) The foregoing outlines features of several embodiments so that those skilled in the art may better understand aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for manufacturing an integrated circuit device, comprising: (a) providing a photonic structure including an insulating structure and an optical coupler embedded in the insulating structure; and (b) removing a portion of the insulating structure to expose a coupling surface of the optical coupler and form a light reflective structure corresponding to the coupling surface; wherein (b) comprises: (b1) removing a first portion of the insulating structure to expose the coupling surface of the optical coupler; (b2) removing a second portion of the insulating structure to form a rib portion on the insulating structure; and (b3) shaping the rib portion to form the light reflective structure.

2. The method of claim 1, wherein the light reflective structure includes an inclined reflective surface, and (b3) comprises: (b31) hitting a portion of the rib portion to form the inclined reflective surface by plasma.
3. The method of claim 2, wherein after (b), the method further comprises: (c) forming a light reflective film on the inclined reflective surface.
4. The method of claim 3, wherein the light reflective film is a distributed Bragg reflector (DBR) film.
5. The method of claim 2, wherein in (b31), the portion of the rib portion is a corner portion.
6. The method of claim 2, wherein an angle between the inclined reflective surface and a lateral surface of the rib portion is greater than 90 degrees.
7. The method of claim 1, wherein the light reflective structure includes an inclined reflective surface, and the inclined reflective surface is lower than the optical coupler.
8. The method of claim 1, wherein the light reflective structure includes an inclined reflective surface, and the inclined reflective surface is lower than a top surface of the optical coupler.
9. The method of claim 1, wherein (a) includes: (a1) forming an upper light reflector in the insulating structure and above the optical coupler.
10. The method of claim 1, wherein (a) includes: (a1) forming a lower light reflector in the insulating structure and below the optical coupler.
11. A method for manufacturing an integrated circuit device, comprising: (a) providing a photonic structure including an insulating structure and at least two optical couplers embedded in the insulating structure and disposed side by side; and (b) removing a portion of the insulating structure to form a light reflective structure between the at least two optical couplers; wherein (b) comprises: (b1) removing a first portion of the insulating structure to expose coupling surfaces of the optical couplers; (b2) removing a second portion of the insulating structure to form a rib portion between the optical couplers; and (b3) shaping the rib portion to form the light reflective structure.
12. The method of claim 11, wherein the at least two optical couplers include a first optical coupler and a second optical coupler spaced apart from the first optical coupler, and the light reflective structure includes a first inclined reflective surface corresponding to the first optical coupler and a second inclined reflective surface corresponding to the second optical coupler.
13. The method of claim 12, wherein (b3) comprises: (b31) hitting a first portion of the rib portion to form the first inclined reflective surface and hitting a second portion of the rib portion to form the second inclined reflective surface by plasma.
14. The method of claim 13, wherein an inclined direction of the first inclined reflective surface is opposite to an inclined direction of the second inclined reflective surface.
15. The method of claim 13, wherein after (b), the method further comprises: (c) forming a light reflective film on the first inclined reflective surface and the second inclined reflective surface.
16. The method of claim 12, wherein the first inclined reflective surface is lower than a top surface of the first optical coupler.
17. A method for manufacturing an integrated circuit device, comprising: (a) providing a photonic structure including an insulating structure and an optical coupler embedded in the insulating structure, wherein (a) includes: (a1) forming an upper light reflector in the insulating structure and above the optical coupler; (b) bonding the photonic structure and an electronic structure together; and (c) removing a portion of the insulating structure of the photonic structure to expose a coupling surface of the optical coupler and form a light reflective structure corresponding to the coupling surface.
18. The method of claim 17, wherein the light reflective structure includes an inclined reflective surface.
19. The method of claim 18, further comprising forming a light reflective film on the inclined reflective surface.

20. The method of claim 17, wherein (a) further includes: (a2) forming a lower light reflector in the insulating structure and below the optical coupler.